



Data Mining: Concepts and Techniques

Why Data Mining?



- **The Explosive Growth of Data: from terabytes to petabytes**
 - **Data collection and data availability**
 - **Automated data collection tools, database systems, Web, computerized society**
 - **Major sources of abundant data**
 - **Business: Web, e-commerce, transactions, stocks, ...**
 - **Science: Remote sensing, bioinformatics, scientific simulation, ...**
 - **Society and everyone: news, digital cameras, YouTube**
- **We are drowning in data, but starving for knowledge!**
- **“Necessity is the mother of invention”—Data mining—
Automated analysis of massive data sets**

Evolution of Sciences

- Before 1600, empirical science
- 1600-1950s, theoretical science
 - Each discipline has grown a *theoretical* component. Theoretical models often motivate experiments and generalize our understanding.
- 1950s-1990s, computational science
 - Over the last 50 years, most disciplines have grown a third, *computational* branch (e.g. empirical, theoretical, and computational ecology, or physics, or linguistics.)
 - Computational Science traditionally meant simulation. It grew out of our inability to find closed-form solutions for complex mathematical models.
- 1990-now, data science
 - The flood of data from new scientific instruments and simulations
 - The ability to economically store and manage petabytes of data online
 - The Internet and computing Grid that makes all these archives universally accessible
 - Scientific info. management, acquisition, organization, query, and visualization tasks scale almost linearly with data volumes. **Data mining** is a major new challenge!
- Jim Gray and Alex Szalay, *The World Wide Telescope: An Archetype for Online*

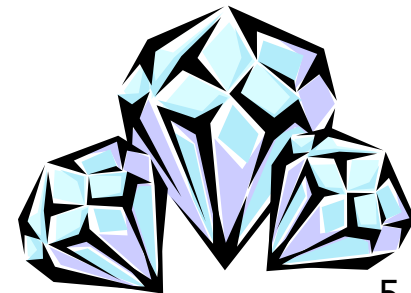
Evolution of Database Technology

- **1960s:**
 - Data collection, database creation, IMS and network DBMS
- **1970s:**
 - Relational data model, relational DBMS implementation
- **1980s:**
 - RDBMS, advanced data models (extended-relational, OO, deductive, etc.)
 - Application-oriented DBMS (spatial, scientific, engineering, etc.)
- **1990s:**
 - Data mining, data warehousing, multimedia databases, and Web databases
- **2000s**
 - Stream data management and mining
 - Data mining and its applications
 - Web technology (XML, data integration) and global information systems

What Is Data Mining?

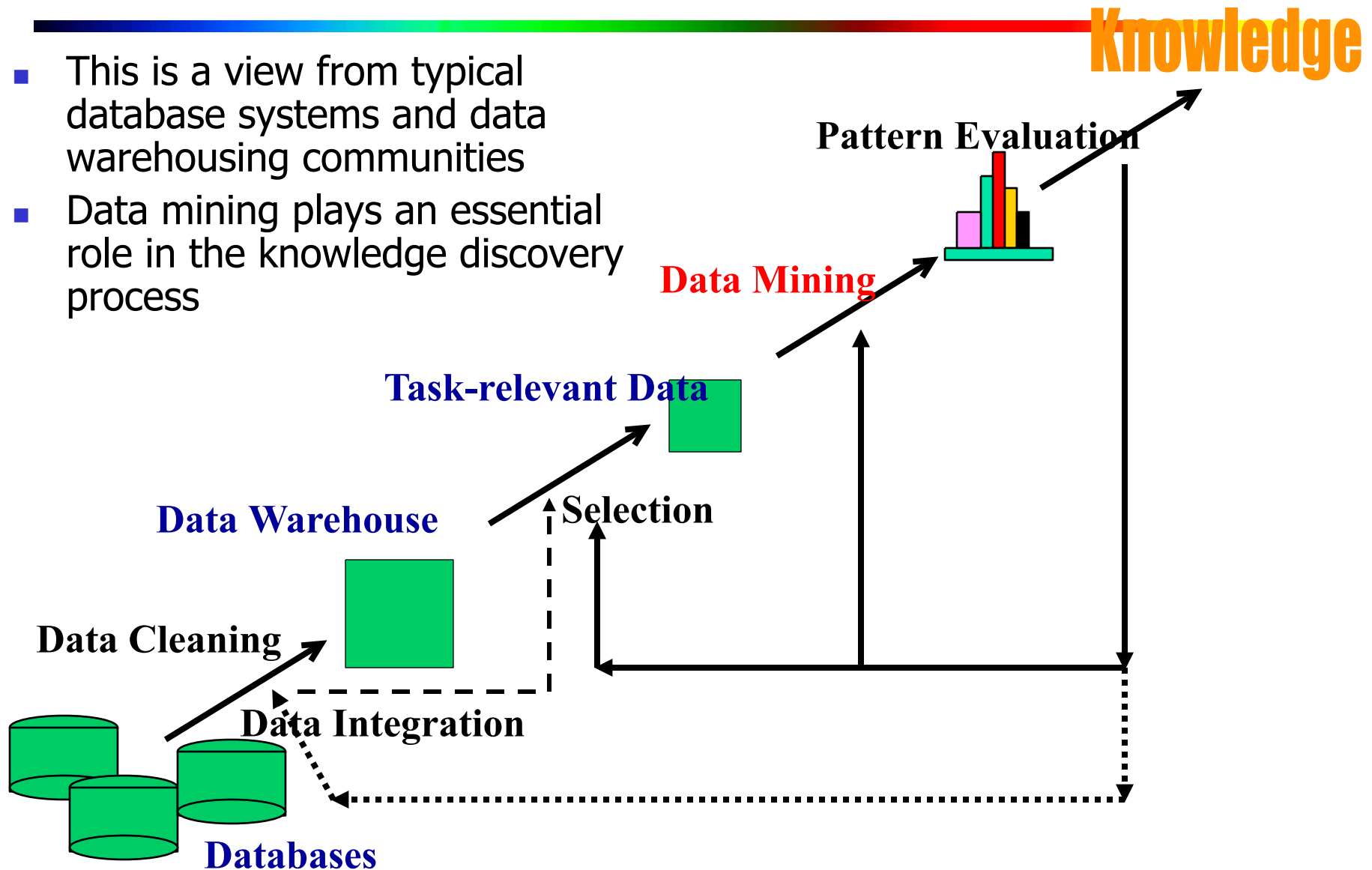


- **Data mining (knowledge discovery from data)**
 - Extraction of interesting (non-trivial, implicit, previously unknown and potentially useful) patterns or knowledge from huge amount of data
 - Data mining: a misnomer?
- **Alternative names**
 - Knowledge discovery (mining) in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, data dredging, information harvesting, business intelligence, etc.
- **Watch out: Is everything “data mining”?**
 - Simple search and query processing
 - (Deductive) expert systems

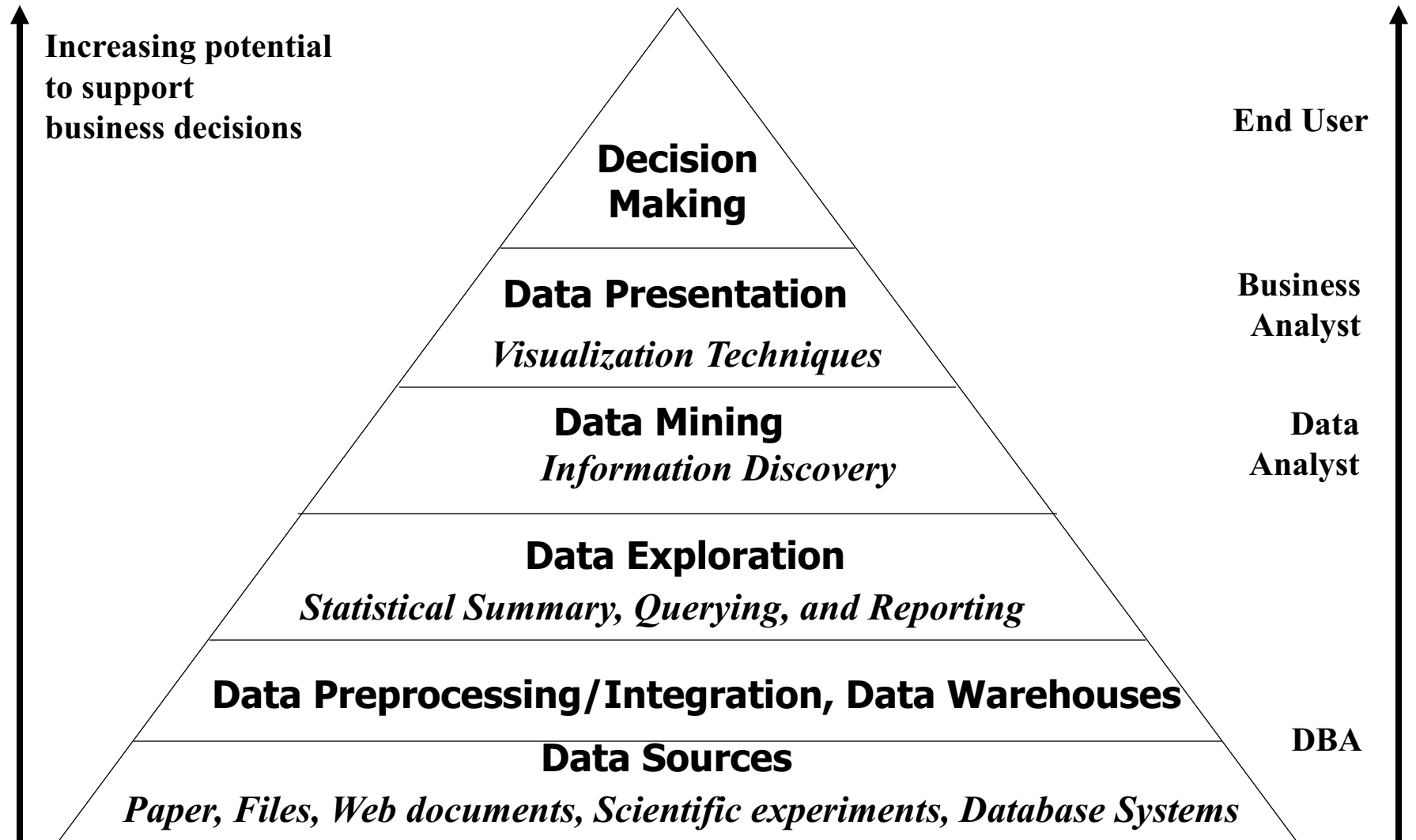


Knowledge Discovery (KDD) Process

- This is a view from typical database systems and data warehousing communities
- Data mining plays an essential role in the knowledge discovery process



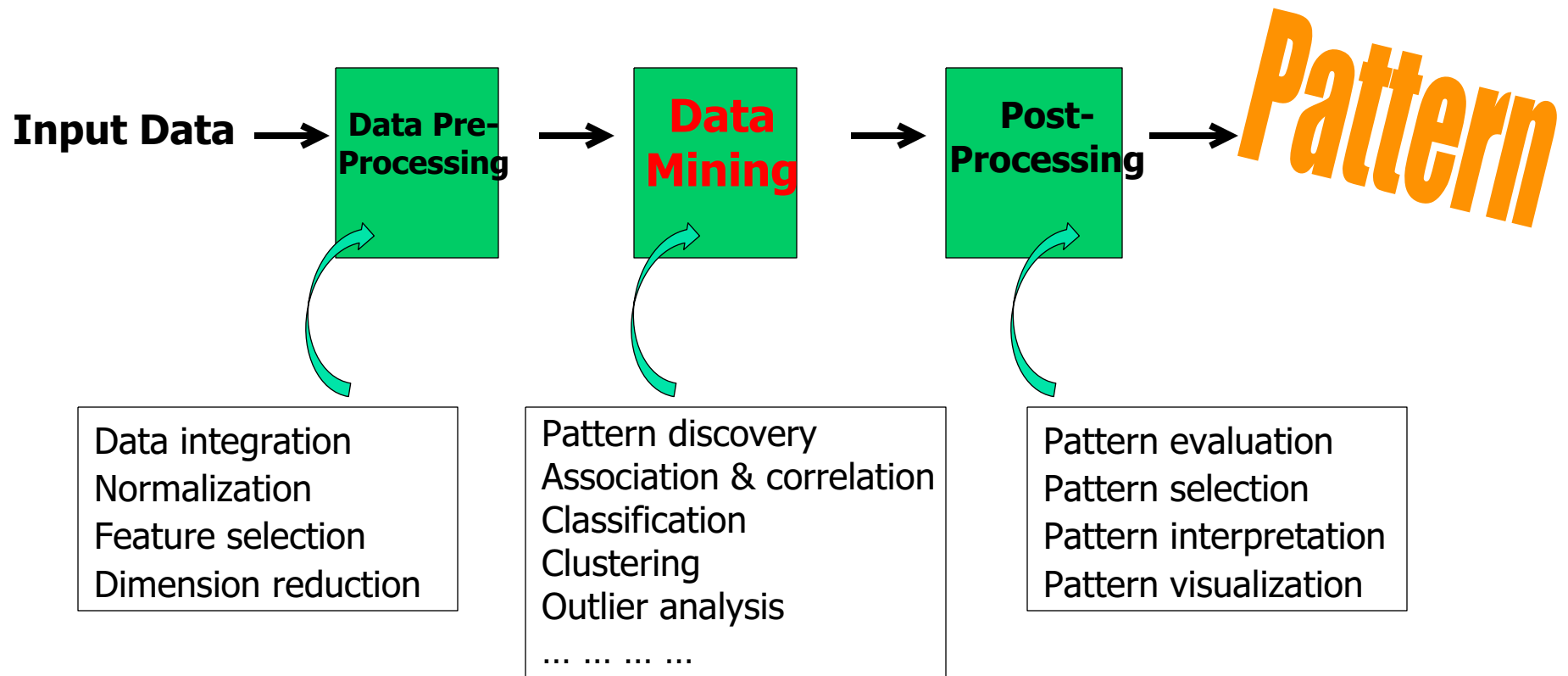
Data Mining in Business Intelligence



Example: Mining vs. Data Exploration

- Business intelligence view
 - Warehouse, data cube, reporting but not much mining
- Business objects vs. data mining tools
- Supply chain example: tools
- Data presentation
- Exploration

KDD Process: A Typical View from ML and Statistics



- This is a view from typical machine learning and statistics communities

Example: Medical Data Mining

- Health care & medical data mining – often adopted such a view in statistics and machine learning
- Preprocessing of the data (including feature extraction and dimension reduction)
- Classification or/and clustering processes
- Post-processing for presentation

Multi-Dimensional View of Data Mining

■ Data to be mined

- Database data (extended-relational, object-oriented, heterogeneous, legacy), data warehouse, transactional data, stream, spatiotemporal, time-series, sequence, text and web, multi-media, graphs & social and information networks

■ Knowledge to be mined (or: Data mining functions)

- Characterization, discrimination, association, classification, clustering, trend/deviation, outlier analysis, etc.
- Descriptive vs. predictive data mining
- Multiple/integrated functions and mining at multiple levels

■ Techniques utilized

- Data-intensive, data warehouse (OLAP), machine learning, statistics, pattern recognition, visualization, high-performance, etc.

■ Applications adapted

- Retail, telecommunication, banking, fraud analysis, bio-data mining, stock market analysis, text mining, Web mining, etc.

Data Mining: On What Kinds of Data?

- Database-oriented data sets and applications
 - Relational database, data warehouse, transactional database
- Advanced data sets and advanced applications
 - Data streams and sensor data
 - Time-series data, temporal data, sequence data (incl. bio-sequences)
 - Structure data, graphs, social networks and multi-linked data
 - Object-relational databases
 - Heterogeneous databases and legacy databases
 - Spatial data and spatiotemporal data
 - Multimedia database
 - Text databases
 - The World-Wide Web